Airport Announcements with Kokoro

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Kokoro: the live voices

Kokoro makes every announcement live on the player's PC, free, with no cloud service or monthly fee. The game already uses it today.

- **Licence:** Kokoro v1.0 is Apache 2.0, and the code around it (kokoro-onnx, misaki) is MIT. Both allow sale in a paid game with credit. They are already listed in the third-party notices.
- **Quality:** very natural for calm speech, with many male and female voices, accents and ages.
- **Live text:** it can say anything as it happens: a passenger's name, a flight number, a gate change. For example, "Passenger James Hansen, please go to gate 4 immediately, your aircraft is ready to depart."
- **Speed:** a short line takes a fraction of a second to make.
- **Limit:** weak on strong emotions. Angry, shouting or drunk lines are made in advance with Chatterbox instead (MIT licence, to be checked before use).

The loudspeaker sound

The clean Kokoro voice is sent through Unreal's audio effects so it sounds like a real airport PA. All of it works live and costs very little performance.

Effect	What it does	How it sounds
Speaker tone	Cuts the deep bass and the highest treble (roughly 250 Hz to 5 kHz kept)	Thin and boxy, like ceiling speakers
Speaker grit	Light distortion and compression	Amplified, not studio-clean
Hall echo	Reverb shaped per area	Long in the main hall, shorter at the gates, almost none in the jetway
Many speakers	The voice plays from real speaker spots in the ceiling, further ones a fraction later	The layered airport echo, changing as you walk
Chime	Two- or three-tone gong before and after	The classic airport call

Effect	What it does	How it sounds
Ducking	Crowd murmur and shop music dip while the announcer speaks	Announcements stand out
Rooms and outside	Muffled through doors and glass	Distant in the lounge or on the kerb
Your own layer	A custom sound (chime, crackle) mixed on top	Your airport's signature

Step by step in Unreal Engine 5

The announcements are built in ten steps, after the first test of the main hall. Steps 1 to 4 make the voice, 5 to 8 make the loudspeaker sound, 9 and 10 connect it to the game.

1. **Bring the Kokoro model over.** Copy the Kokoro v1.0 voice model and voice files from the Babylon edition into the Unreal project, with their licence texts.
2. **Run it inside the game.** Load the model with Unreal's built-in Neural Network Engine (it runs ONNX models on the player's PC). No internet needed.
3. **Turn text into speech sounds.** Port the misaki word dictionary (MIT) to C++, so names and words are pronounced correctly. Do not use espeak-ng: it is GPL, which the Unreal licence forbids in the game.
4. **Play the voice as a sound.** Feed the generated audio into a procedural sound wave, so a new line can play as soon as it is made.
5. **Make the PA sound chain.** Create a "PA" submix with the speaker tone, grit and compression effects, so every announcement goes through it.
6. **Place the ceiling speakers.** Add speaker points along the terminal ceilings. Each one plays the announcement with a small delay by distance, for the layered echo.
7. **Shape the echo per area.** Add audio volumes with their own reverb: long for the main hall, shorter for the gates, almost none for the jetways, muffled for rooms and outside.
8. **Add the chime, ducking and your layer.** Play the gong before and after, lower the crowd and shop sounds during the call, and mix in your own custom layer.
9. **Write the announcement system.** Templates with slots for name, flight, gate and time (boarding, last call, gate change, delay, name call), triggered by the flight's real state, with a queue so calls never overlap.
10. **Add the settings.** Announcer voice, announcement volume, which kinds of calls are on, and your custom layer on or off. You listen in the terminal and tell me what to change.

As the terminal grows, new halls get their own speaker points and echo, so steps 6 and 7 repeat for each extension.

Licence checklist

Everything here is free and allowed in a paid game, as long as these rules are kept.

- [ ] Kokoro v1.0 (Apache 2.0), kokoro-onnx and misaki (MIT): credit and licence texts in the third-party notices.
- [ ] Chatterbox (MIT) for the emotional lines: licence checked again before use, then listed.
- [ ] No espeak-ng or other GPL parts (forbidden by the Unreal licence).
- [ ] No copying of real people's voices. Reference recordings for drunk or angry delivery only from people who agree to it.
- [ ] Chime and custom layer sounds are our own recordings.
- [ ] Steam store page: tick the disclosure for generated voice lines.

Not used: Coqui XTTS and F5-TTS (non-commercial models), Fish Audio's model (paid licence), cheap one-time voice tools that wrap those models. ReadSpeaker (from $2,000 one-time per game, 4 voices included) stays a later option.